



Sample select scripts explained

1. View all employees

```
SELECT * FROM employees;
```

Meaning:

Fetches all employee records.

2. JOIN: Employees + Departments

```
SELECT
    e. employee_id,
    e.first_name,
    e.last_name,
    d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id;
```

What it does:

Combines:

- Employees table
- Departments table

LEFT JOIN meaning:

- Show ALL employees
- Even if department is missing



3. JOIN: Employees + Roles + Salary

```
SELECT
    e.first_name,
    e.last_name,
    jr.role_title,
    jr.base_salary
```

What it does:

Combines 3 tables:

- employees
- employee_positions
- job_roles

Result:

You see:

- Who the employee is
- Their job role
- Their salary

4. AGGREGATION: Average Salary by Department

```
SELECT
    d.department_name,
    AVG(jr.base_salary) AS average_salary
```

What it does:

Calculates:

Average salary per department

Key concept:

- AVG() = aggregation function
- GROUP BY = groups results per department

Final Output Meaning

This query answers:

“Which departments are the highest paying on average?”



BIG PICTURE (How the System Works)

Your database models a real company:

